Orchestration design

Multi-Agent AI for Smart Contract Audit
and Cybersecurity Risk Assessment

Design objective. Preserve independent evidence, give reviewers one controlled opportunity to reconsider disputed claims, and retain a transparent before/after record.

GPT-6.1 Sol, Claude Opus 5.5 and Grok 4.7 run through Amazon Bedrock. A deterministic TypeScript supervisor manages execution and validates reports. It does not add a fourth adjudicating model or use managed Bedrock Agents.

One baseline. One optional reconsideration.

Stage	Input and responsibility	Retained output
1. Independent review	Same numbered source and prompt variant for all three models. No static findings or peer answers.	Source-bound original findings, model outcomes and independent agreement.
2. Queue selection	Identify severity differences, non-reporting and unavailable coverage. Select the first 12 queued claims in stable order.	Selected IDs, original contributions and explicitly deferred claims.
3. Second review	At least two first-pass models must have succeeded. Those reviewers reconsider the same claims with peer context, including their own.	Support, rejection or uncertainty with rationale and exact source evidence.
4. Separate summary	Validate every assessment and derive a peer-informed summary. Preserve the first-pass snapshot unchanged.	Before/after decisions, missing reviewers, failures and unresolved items.

Why preserve the first pass?

First-pass agreement is independent corroboration among model outputs, not a probability of correctness. Second-pass reviewers have seen peer claims, so their responses are correlated. Their decisions must not be added to the original votes or used to inflate the confidence display.

Scope: the second review reconsiders selected existing claims. It cannot discover a vulnerability missed by every first-pass model, recover a failed reviewer or automatically revisit unanimous unqueued findings.

A bounded decision process

Explicit reviewer decisions, finite requests and separate records for each pass.

Second-pass evidence	Derived summary
All three configured models support a claim and propose the same severity	Supported after peer review
All three explicitly do not support the claim	Not supported after peer review
Mixed decisions, uncertainty, different severities, a missing reviewer or deferred work	Unresolved

Two agreeing reviewers with a missing third remain unresolved. A reviewer may withdraw its own original claim. Neither a supported summary nor a rejected claim is a human-validated security conclusion.

Execution budget

Boundary	Enforced limit
Per model, per pass	At most 2 attempts within one absolute 180-second deadline
Per server request	210 seconds inside a 240-second Vercel function; each pass is a separate request
Second review scope	One round; at most 12 queued claims; only successful first-pass reviewers
Provider requests	At most 6 per pass, or 12 for one contract with both passes
Five-case batch	First only: at most 30 calls / 20 minutes. With second review: 60 calls / 40 minutes.

Selective retry within the deadline

Only explicit HTTP 429, 500, 502, 503 and 504 responses may receive one retry. Bounded backoff stays inside the original deadline. Authentication, permission, invalid evidence, malformed output, network ambiguity and local timeout failures are terminal. A retry never adds an independent vote.

The key stays in tab and active request memory. Cancellation stops new work but cannot reverse AWS work already accepted. Returned usage may omit failed attempts. In-memory guards are per runtime, not a distributed account quota or guaranteed dollar cap.

Evaluate the same evidence

Keep the five existing contracts fixed. Pair every second review with its own original first-pass report.

Enable **Include second review** in five-case validation to collect both stages. Each case runs sequentially; its first-pass record is the baseline for reconsideration. Keep failures, skipped reviews and deferred claims in the export. Run Standard and Guided examples as separate explicit experiments.

Separate operational results from quality

Measure	Evidence required
Execution reliability	Model completion, attempts, errors, timeouts, elapsed time and available usage for each pass
Decision changes	Same candidate ID and baseline; original support versus explicit second-pass support, rejection or uncertainty
False positives	Human judgment that a reported claim is unsupported, followed by review of how the second pass treated it
Missed issues	An independently reviewed issue inventory, including vulnerabilities absent from every model report
Regression risk	Supported first-pass findings incorrectly rejected or obscured after reconsideration

Human review protocol

1. Review each distinct root-cause/location claim before consulting the peer-informed summary. Mark it supported, unsupported or unresolved, with a reason.

2. Record the second-pass judgment separately. Preserve any disagreement between reviewers; do not treat a changed model opinion as ground truth.

3. Compare paired counts and execution outcomes, retaining skipped/deferred cases. A smaller finding count or greater agreement alone does not establish higher accuracy.

The five sources are teaching fixtures with informative comments, not a held-out benchmark. Automated tests verify implementation behavior with simulated provider responses. Live availability and quality improvements require authorized runs and reviewed labels.

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